CompTIA A+ Core 1 Playbook

IT Labs
Prepared by Garrick White

6.5 Install a SOHO Network

You are creating a home office network. The network's internet access is connected to the left network port in the wall switch plate. You have selected a router and a wireless access point to use. Now you need to connect the devices and configure the network connections.

In this lab, your task is to:

- Connect the router to the left Ethernet port in the wall plate and provide power to the router.
- Connect both desktop computers to the router.
- Configure both desktop computers to receive DHCP addresses and be assigned DNS servers automatically.
- Connect the wireless access point to the router.
  - Use **192.168.0.100** to connect to the wireless access point management interface.
    - Username: **admin**
    - Password: **password**
- Configure the wireless access point using Chrome on PC1.
  - SSID: **HomeWireless**
  - Authentication Mode: **WPA2-PSK**
  - Encryption Method: **AES**
  - Pass Phrase: **@hOMEwIRELESS!**
- Configure the laptop to connect to the wireless network.
- Configure the iPad to connect to the wireless network.

Explanation

Complete this lab as follows:

1. Connect the router to the internet.
    1. From the home office view, under *Computer Desk*, select **Hardware**.
    2. Under Inventory, expand **Networking Devices**.
    3. Drag the **router** from the Inventory to the Workspace area.
    4. Switch to the back view of the router.
    5. Under Inventory, expand **Cables**.
    6. Select the **Power Adapter**.
    7. From the Selected Item pane:
        - Drag the **DC Power Connector** to the power plug on the router.
        - Drag the **AC Power Adapter** to the AC power plug on the surge protector.
    8. Under Inventory, select **Cat5e Cable, RJ45**.
    9. From the Selected Item pane:
        - Drag an **RJ45 Connector** to the WAN port on the router.
        - Drag the other **RJ45 Connector** to the left network plug on the wall plate.
2. Connect the desktop computers to the router.
    1. Switch to the Back view of the PC1 case (on the left).
    2. From the Inventory, select **Cat5e Cable, RJ45**.
    3. From the Selected Item pane:
        - Drag an **RJ45 Connector** to a LAN port on the router.
        - Drag the other **RJ45 Connector** to the network plug on the computer.
    4. Repeat these steps to connect PC2 to the router.
3. Configure TCP/IP on the desktop computers.
    1. Click the PC1 monitor to view the OS.
    2. Right-click **Start** and then select **Settings**.
    3. Select **Network & internet**.
    4. Select **Ethernet**.
    5. From the right pane, scroll down to *IP assignment* and then select **Edit**.
    6. Under *Edit IP settings*, use the drop-down list to select **Automatic (DHCP)**.
    7. Select **Save**.
    8. From the left pane of the Settings app, select **Network & internet** and verify that the computer has access to the internet.

    9. From the top left, select **Computer Desk** to switch back to the Workspace view.
    10. On the PC2 monitor, select **Click to view Windows 11**.
    11. Repeat these steps to configure PC2.
4. Connect the wireless access point to the router.
    1. From the top left, select **Computer Desk** to return to the Workspace.
    2. Under Inventory, expand **Networking Devices**.
    3. Drag the **wireless access point** from the Inventory to the Workspace area.
    4. Switch to the **Back** view of the wireless access point.
    5. Under Inventory, expand **Cables**.
    6. Select the **Power Adapter**.
    7. From the Selected Item pane:
        - Drag the **DC Power Connector** to the power plug on the wireless access point.
        - Drag the **AC Power Adapter** to the AC power plug on the surge protector.
    8. Under Inventory, select the **Cat5e Cable**.
    9. From the Selected Item pane:
        - Drag an **RJ45 Connector** to the WAN port on the wireless access point.
        - Drag the other **RJ45 Connector** to a LAN port on the router.
5. Configure the wireless access point.
    1. Click the PC1 monitor to view the OS.
    2. From the taskbar, select **Google Chrome**.
    3. In the URL field, enter **192.168.0.100** and press **Enter** to view the wireless access point management console.
    4. In the Windows Security pop-up dialog, enter **admin** for the username.
    5. Enter **password** in the Password field.
    6. Select **OK**.
    7. From the left menu, select **Wireless**.
    8. Select **Basic**.
    9. Enter **HomeWireless** in the *Wireless Name(SSID)* field.
    10. Select **Apply**.
    11. In the Windows Security pop-up dialog, enter **admin** for the username.
    12. Enter **password** in the Password field.
    13. Select **OK**.
    14. Under Wireless, select **Security**.

15. From the Security Mode drop-down list, select **WPA2-PSK** for authentication.
16. Under WPA, make sure **AES** is selected.
17. In the Pass Phrase field, enter **@hOMEwIRELESS!** as the security key.
18. Select **Apply**.

6. Configure the laptop's wireless connection.
    1. From the top left, select **Computer Desk** to switch back to the Workspace view.
    2. On the front of the laptop, select the **wireless switch** to enable wireless networking.
    3. Click the laptop monitor to view the OS.
    4. In the system tray, select the **network icon**.
    5. Select the **arrow** above Available**.**
    6. Select **HomeWireless**.
    7. Make sure **Connect Automatically** is checked.
    8. Select **Connect**.
    9. Enter **@hOMEwIRELESS!** as the network security key.
    10. Select **Next**.
    11. Using your mouse, hover over the *networking icon* and verify that the laptop has a connection to the internet.
7. Configure the iPad wireless connection.
    1. From the top left, select **Computer Desk** to switch back to the Workspace view.
    2. Click the button at the bottom of the iPad to turn it on and view the OS.
    3. Select **Settings**.
    4. Select **Wi-Fi**.
    5. Under NETWORKS, select **HomeWireless**.
    6. In the Password field, enter **@hOMEwIRELESS!** as the password.
    7. Select **Join**.